W2 ELK Stack + Log Pipeline

Elasticsearch | Logstash | Kibana | Filebeat | Day 6-10

5 DAYS

Project Information

Project	Healthcare IoT Deception Honeypot Network
Week	Week 2 of 4 — ELK Stack Setup
Services	Elasticsearch 8.6 + Logstash + Kibana + Filebeat
Ports	ES: 9200 | Kibana: 5601 | Logstash: 5044
Data Flow	Cowrie Logs → Filebeat → Logstash → Elasticsearch → Kibana
Stack	Python + Flask + Docker + ELK Stack

Week 2 Objectives

- Download MaxMind GeoLite2 City database for IP geolocation
- Write 3 Logstash pipeline files: input (with beats port 5044), filter, output
- Configure Filebeat to ship Cowrie JSON logs to Logstash
- Launch full ELK stack and verify all 5 containers running
- Create Kibana index pattern honeypot-* and view attack events

DAY 6 GeoIP Database Setup

STEP 1 Download MaxMind GeoLite2

Goal: Free IP geolocation database

```
https://dev.maxmind.com/geoip/geolite2-free-geolocation-data
```

The first command extracts the downloaded GeoLite2 compressed file into the Downloads directory. The -xzf option is used to extract a .tar.gz file, and -C specifies the destination folder. The second command then moves the GeoLite2 database file into the project's data folder so that it can be used by Logstash for IP geolocation.

```
tar -xzf ~/Downloads/GeoLite2-City_*.tar.gz -C ~/Downloads/

mv ~/Downloads/GeoLite2-City_*/GeoLite2-City.mmdb ./data/GeoLite2-City.mmdb
```

This command verifies that the database file exists and displays its size in a human-readable format. Expected output should show approximately 65MB for the GeoLite2-City.mmdb file.

```
ls -lh data/GeoLite2-City.mmdb
```

Expected Output:

```
-rw-r--r-- 1 user user 65M ... data/GeoLite2-City.mmdb
```

DAY 7 Logstash Pipeline Files

STEP 2 Create 01_input.conf

Goal: Input with BOTH beats AND file

This command creates a Logstash input configuration file. The beats section enables Logstash to receive logs from Filebeat on port 5044. The file section tells Logstash to read JSON log files generated by the Cowrie honeypot directly. The start_position beginning ensures logs are read from the start, sincedb_path /dev/null disables tracking so logs are always reprocessed, and codec json instructs Logstash to parse incoming data in JSON format. NOTE: The beats section is mandatory -- without it, Filebeat will not be able to connect to Logstash.

```
cat > logging/logstash/pipeline/01_input.conf << 'EOF'
input {
  beats {
    port => 5044
  }
  file {
    path => "/var/log/honeypot/cowrie/*.json"
    start_position => "beginning"
    sincedb_path => "/dev/null"
    codec => json
```

```
    type => "cowrie"
  }
}
EOF
```

NOTE: beats section zaruri hai — iske bina Filebeat connect nahi hoga!

STEP 3	Create 02_filter.conf + 03_output.conf	**Goal:** Filter + Output

This command creates the filter configuration file used to process and enrich log data. The date filter converts the timestamp field into a proper date format for accurate time-based analysis. The geoip filter adds geographical information like country and city based on the source IP address using the GeoLite2 database. The mutate filter adds a custom field called attack_type to classify different types of attacks such as brute force, credential theft, command injection, or malware delivery.

```
cat > logging/logstash/pipeline/02_filter.conf << 'EOF'
filter {
  if [type] == "cowrie" {
    date { match => ["timestamp", "ISO8601"] target => "@timestamp" }
    geoip { source => "src_ip" target => "geoip" database =>
"/usr/share/logstash/GeoLite2-City.mmdb" }
    if [eventid] == "cowrie.login.success" {
      mutate { add_field => { "attack_type" => "credential_theft" } }
    } else if [eventid] == "cowrie.login.failed" {
      mutate { add_field => { "attack_type" => "bruteforce" } }
    } else if [eventid] == "cowrie.command.input" {
      mutate { add_field => { "attack_type" => "command_injection" } }
    } else if [eventid] == "cowrie.session.file_download" {
      mutate { add_field => { "attack_type" => "malware_delivery" } }
    }
  }
}
EOF
```

This command creates the output configuration file that defines where processed logs will be sent. The elasticsearch block sends all processed logs from Logstash to the Elasticsearch service running on port 9200. The index setting creates daily indices in Elasticsearch using the date pattern, helping organize and search logs by date efficiently.

```
cat > logging/logstash/pipeline/03_output.conf << 'EOF'
output {
  elasticsearch {
    hosts => ["elasticsearch:9200"]
    index => "honeypot-%{+YYYY.MM.dd}"
  }
```

```
}
EOF
```

DAY 8 Filebeat Configuration — Important Fix

STEP 4 Create filebeat.yml with Correct Ownership

Goal: Filebeat ownership fix required

This command creates the Filebeat configuration file with root privileges using sudo tee. The paths section tells Filebeat where to find the honeypot log files that need to be collected. The json.keys_under_root: true setting ensures JSON fields are stored at the top level instead of being nested. The output.logstash section configures Filebeat to send collected logs to Logstash on port 5044.

```
sudo tee logging/filebeat/filebeat.yml << 'EOF'
filebeat.inputs:
  - type: log
    enabled: true
    paths:
      - /var/log/honeypot/cowrie/*.json
    json.keys_under_root: true
    json.overwrite_keys: true
    tags: ["cowrie"]

output.logstash:
  hosts: ["LOGSTASH_IP:5044"]
EOF
```

These two commands are required for Filebeat to start correctly. The chown command changes ownership of the Filebeat configuration file to root, and the chmod command removes write permissions for group and others to prevent unauthorized modifications. Filebeat will refuse to run if its config file has incorrect permissions.

```
sudo chown root:root logging/filebeat/filebeat.yml
sudo chmod go-w logging/filebeat/filebeat.yml
```

NOTE: LOGSTASH_IP = docker inspect logstash | grep IPAddress | grep 192

DAY 9 Full ELK Stack Launch

STEP 5 Start All ELK Services

Goal: Launch everything

The docker compose up -d command starts all ELK stack services -- Elasticsearch, Kibana, Logstash, and Filebeat -- in detached mode, allowing them to run in the background. The docker

compose ps command then displays the status of all running containers to confirm all services are active.

```
docker compose up -d elasticsearch kibana logstash filebeat

docker compose ps
```

This loop repeatedly checks whether Elasticsearch is fully started and ready to accept requests. It sends a silent HTTP request every 3 seconds until Elasticsearch responds successfully, ensuring other services do not fail due to Elasticsearch not being ready yet.

```
until curl -sf http://localhost:9200 > /dev/null; do

  echo -n '.'; sleep 3

done

echo ' Ready!'
```

This command lists all indices in Elasticsearch to verify that logs are being stored correctly. After successful data flow, a honeypot-* index with today's date should appear in the output.

```
curl http://localhost:9200/_cat/indices?v
```

Expected Output:

```
 Ready!
honeypot-2026.03.21 open
```

NOTE: First time: 10-15 min for image download

DAY 10 Kibana Setup

STEP 6 Create Kibana Index Pattern

Goal: Setup visualization

Open Kibana at http://localhost:5601 in your browser. Navigate to Stack Management, then Index Patterns, and create a new index pattern using honeypot-*. Select the @timestamp field as the time filter. Then go to the Discover tab to view and explore all captured attack events visually.

```
1. http://localhost:5601 kholo

2. Explore on my own click karo

3. Stack Management -> Index Patterns -> Create

4. Pattern: honeypot-*  |  Timestamp: @timestamp

5. Save -> Discover tab mein events dikhenge
```

This command counts the total number of log entries stored in all Elasticsearch honeypot indices. A non-zero count confirms that logs from the Cowrie honeypot are successfully flowing through Filebeat and Logstash into Elasticsearch.

```
curl -s http://localhost:9200/honeypot-*/_count
```

Expected Output:

```
{"count":18}
```

STEP 7

Filebeat Connection Fix (If Needed)

Goal:
Fix Filebeat-Logstash connection

This command retrieves the IP address assigned to the Logstash container on the honeypot Docker network. This IP is needed to update the Filebeat configuration so it can correctly reach Logstash.

```
docker inspect logstash | grep '"IPAddress"'
```

This recreates the Filebeat configuration file with the correct Logstash IP address obtained from the previous step. Using the actual container IP instead of a hostname ensures reliable communication between Filebeat and Logstash on the Docker network.

```
sudo tee logging/filebeat/filebeat.yml << 'EOF'
filebeat.inputs:
  - type: log
    enabled: true
    paths:
      - /var/log/honeypot/cowrie/*.json
    json.keys_under_root: true
    json.overwrite_keys: true
    tags: ["cowrie"]

output.logstash:
  hosts: ["192.168.100.3:5044"]
EOF
```

After updating the config, these commands fix the file ownership and permissions as required by Filebeat, then restart the Filebeat container to apply the new configuration. The logs command then verifies that Filebeat has successfully connected to Logstash at the specified IP and port.

```
sudo chown root:root logging/filebeat/filebeat.yml
sudo chmod go-w logging/filebeat/filebeat.yml
docker compose restart filebeat
sleep 10
docker compose logs filebeat | grep -i connect | tail -3
```

Expected Output:

```
Connecting to backoff(async(tcp://192.168.100.3:5044))
```

NOTE: 192.168.100.3 ki jagah actual Logstash honeypot_net IP daalo

Completion Checklist

- ✓ GeoLite2-City.mmdb exists in data/ folder (65MB)
- ✓ 01_input.conf has BOTH beats {port=>5044} AND file input
- ✓ 02_filter.conf classifies all 4 attack types correctly
- ✓ filebeat.yml owned by root (sudo chown root:root)
- ✓ All 5 containers Up in docker compose ps
- ✓ curl http://localhost:9200/_cat/indices shows honeypot-* index
- ✓ Kibana opens at http://localhost:5601
- ✓ Attack events visible in Kibana Discover tab

Week 2 — Required GitHub Commits

Commit Message	Files Changed
`git commit -m "feat: add Logstash input pipeline with beats port 5044"`	`logging/logstash/pipeline/01_input.conf`
`git commit -m "feat: add Logstash filter with GeoIP and attack classification"`	`logging/logstash/pipeline/02_filter.conf`
`git commit -m "feat: add Logstash output to Elasticsearch"`	`logging/logstash/pipeline/03_output.conf`
`git commit -m "feat: add Filebeat config with root ownership fix"`	`logging/filebeat/filebeat.yml`
`git commit -m "feat: add ELK services to docker-compose.yml"`	`docker-compose.yml`
`git commit -m "feat: add GeoLite2-City database for IP geolocation"`	`data/GeoLite2-City.mmdb`
`git commit -m "fix: add beats input to logstash 01_input.conf"`	`logging/logstash/pipeline/01_input.conf`
`git commit -m "fix: filebeat ownership and logstash IP fix"`	`logging/filebeat/filebeat.yml`
`git commit -m "test: verify honeypot-* index in Elasticsearch"`	`curl http://localhost:9200/_cat/indices`
`git commit -m "test: verify attack events in Kibana Discover"`	`Kibana index pattern honeypot-*`